NPHD9042 - Applied Multivariate Analysis

Syllabus Download

General Course Information

Course: NPHD9042 - Applied Multivariate Analysis

Term: Fall 2026

Class meeting time and location: Procter 281N or Microsoft Teams via calendar invitation

- 9:30-10:20: Asynchronous video lecture
- 10:30-12:20: Discussion and lab time, either in person or online via Teams

Credit hours/contact hours:

Type	Credit Hours	Clock Hours
Course	3	45

Instructor: Joshua Lambert, PhD, MS, MS

Email: Joshua.lambert@uc.edu

Office hours: By appointment. Please email to schedule.

Communication Policy

Email is the best way to reach me. Please allow at least 24 hours for a response. Emails sent on Fridays will be answered no later than Monday. If you do not hear from me within two business days, please email me again.

Course Description

This course is the second of two courses that introduce advanced statistical methods in doctoral-level nursing research. Multivariate method selection, application, results interpretation, and results presentation are stressed in a flipped-classroom/workshop format. Methods introduced include MANOVA, MANCOVA, factor analysis, principal components analysis, and structural equation modeling. An introduction to power and sample size calculation by method is integrated throughout.

Prerequisite definition: To take this course, students must have taken NPHD9040 with a minimum grade of B- and be enrolled in the NRES-PHD plan.

Classroom Procedures and Policies

All classroom procedures and policies, including required technology, communication policy, netiquette, technology support, accessibility policy, academic integrity, academic honor pledge, religious accommodations, student resources, and student assault resources, can be found on the Canvas course home page under the Classroom Procedures and Policies link.

Student Learning Outcomes

At the end of this course, the student will:

SLO	Student Learning Outcome
SLO #1	Select the appropriate multivariate analysis method based on research objectives and types of variables to be analyzed.
SLO #2	Generate required sample size and power for prototypical analyses by multivariate method, concurrently addressing inclusion of under-represented groups in research.
SLO #3	Display an introductory level ability to manage data and conduct preliminary analyses to assess and address data quality and potential violations of statistical assumptions by multivariate method.
SLO #4	Demonstrate an introductory level ability to generate results using R and interpret relevant results generated for each method.

Pre- and Co-requisites

Pre-requisites: NPHD9000 and NPHD9040 or equivalent.

Co-requisites: None.

Course Format

All students are required to register for Canvas. You are expected to check the course Canvas site and course website regularly:

https://joshuawlambert.github.io/NPHD9042/

You will need access to Microsoft files on the device that you use for the course. If you do not have access to this software on your home device, you can purchase this software at the UC bookstore or use the computers at Procter Hall.

You are required to check your UC email account regularly. The faculty will contact you via your UC email account as needed and will not use a personal, non-UC email account.

Teaching Strategies

Teaching strategies include data analysis and interpretation group workshops, class quizzes, and peer review.

Active learning strategies will be used throughout the semester. Examples include faculty- and student-led discussions, short essay papers, creating professional academic documents, and presentations.

The textbook readings and additional resources posted in the modules provide background information so you can explore, learn about, discuss, and debate the use of statistical analysis in research.

Grammar, spelling, and correct statistical formatting are critical components of scholarly written communication and all professional communication. Please include references when referring to the work of others.

Canvas will be used to submit assignments and communicate. Email will also be used to communicate. The course website will be used to distribute course materials, including lecture notes, videos, homework, projects, and other assignments/materials.

Teaching: Course Philosophy

In my classroom, I view teaching as a partnership between professor and student, grounded in mutual respect, curiosity, and real-world relevance. I strive to connect personally with each learner, recognizing that life circumstances shape how they engage with material, and to present content through verbal, visual, and written modalities so every learning style is addressed.

For PhD nursing students, my focus is on experiential learning: applying statistical methods directly to their own research questions, interpreting results in a supportive, non-competitive environment, and using technology to model the data-driven decisions that shape modern nursing practice. By weaving these elements together, I aim to equip students with both confidence and competence in biostatistics so they can advance science and improve patient care.

To Be Successful

- Get familiar with the course website, video lectures, and other materials.
- Ask questions promptly.
- Engage with the material.
- Familiarize yourself with R and RStudio.
- Meet deadlines consistently.
- Use office hours and resources.
- Participate actively online.
- Apply concepts to your research.

Course Materials

Technology Requirements

- Computer, PC or Mac
- Access to the internet
- Course website: https://joshuawlambert.github.io/NPHD9042/
- Microsoft Office, including Word, Excel, and PowerPoint
- R and RStudio
- University email address
- Canvas

Required Resources

R and RStudio will be used for course examples, assignments, and lab learning.

Optional Resources

Many possible resources are available through the course website:

https://joshuawlambert.github.io/NPHD9042/

Assessments, Assignments, and Grading Policies

Method for Calculation of Course Grade

Workshop Attendance and Participation: 30%

There are 14 class workshops.

Homework: 30%

There are four homework assignments. All homework will be given exactly two weeks prior to its due date. Homework is meant to assess your understanding of statistical software, conceptual knowledge of methods discussed, data quality issues, and statistical interpretation. Each homework assignment will have a more detailed outline of how points are distributed.

Homework Component	Course Grade
Methods	10%
Software	10%
Data quality and statistical interpretation	10%
Total	**30%**

Project Proposals: 10%

Students will complete two proposals: one midterm proposal and one final proposal. Students will choose a dataset of their liking. Dr. Lambert will provide up to three datasets, but students can choose their own. The student should set up a meeting with Dr. Lambert to discuss the project and planned course of action. Each project will require a proposal. The proposal should be 1-2 pages and should mimic a statistical analysis plan.

Proposal Component	Course Grade
Meeting with Dr. Lambert	2%
Statistical analysis plan	6%
Goals	2%
Total	**10%**

Project Papers: 20%

Students will complete two papers: one midterm paper and one final paper. Each project will require a 3-5 page paper. The student's paper is meant to outline the hypothesis of interest, dataset, statistical methods, results, and conclusions.

Paper Component	Course Grade
Hypothesis	4%
Dataset description	4%
Statistical methods	4%
Results	4%

Paper Component	Course Grade
Conclusion	4%
Total	**20%**

Project Presentations: 10%

Students will complete two presentations: one midterm presentation and one final presentation. Each project will include a PowerPoint slide presentation or poster presentation. Creativity is an important aspect of the presentation. All five parts mentioned in the paper grading section should be included with details. Presentations should mimic a conference speed or poster presentation. Presentations may be in person, online synchronous, or online asynchronous. Presentations should be around 10 minutes with 5 minutes of questions.

Presentation Component	Course Grade
Creativity	2%
Project paper components	5%
Questions	3%
Total	**10%**

Grade Breakdown by Group

Assignment Group	Course Grade
Workshop attendance and participation	30%
Homework	30%
Project proposals	10%
Project papers	20%
Project presentations	10%

Grading Scale for Final Grade

Failure to achieve at least 80% (B-) in the course constitutes failure for purposes of the College of Nursing, and the course will have to be retaken.

Letter Grade	Percentage
A	93-100%
A-	90-92.9%
B+	87-89.9%
B	83-86.9%
B-	80-82.9%
C+	77-79.9%
C	73-76.9%
C-	70-72.9%
D+	67-69.9%
D	63-66.9%
D-	60-62.9%
F	<60%

Late Work Policy

Assignments are due by midnight on the date indicated in the course calendar unless otherwise stated. Midnight is based on Eastern Daylight Time/Eastern Standard Time as applicable. This time references the local time in Cincinnati for the duration of the course.

If you anticipate a conflict with any of the due dates, please contact the faculty by email to negotiate any changes. Assignments submitted late without prior written approval will receive a score of 0%.

Grade Review

Any question or discussion over earned points for an examination and/or assignment should be addressed by appealing in writing by email to the course faculty. Refer to the Student Handbook for additional policies regarding grading.

Withdrawal Policy

Students who wish to withdraw from the course are expected to notify the instructor prior to withdrawal. Students can view withdrawal dates on UC's Academic Calendars site. Students who have satisfactorily completed course requirements up to the date of withdrawal will be eligible for a "W" grade; however, it is the instructor’s right to change the “W” to an “F” if it is determined to be warranted through the final grading process. Be sure to refer to UC web registration information regarding other areas that may be affected by the course.

Classroom Procedures/Policies

Technology Support

Please review by accessing the link in the Classroom Procedures, Policies, and Student Resources module in the Canvas course.

Accessibility Policy

Please review by accessing the link in the Classroom Procedures, Policies, and Student Resources module in the Canvas course.

Academic Integrity

Please review by accessing the link in the Classroom Procedures, Policies, and Student Resources module in the Canvas course.

Academic Honor Pledge

Please review by accessing the link in the Classroom Procedures, Policies, and Student Resources module in the Canvas course.

AI Course Policy

Please review by accessing the link in the Classroom Procedures, Policies, and Student Resources module in the Canvas course.

Use generative AI only as a learning aid, not a substitute for your own work. All work submitted must be authored by the student; AI can assist with explanations, brainstorming, or code snippets, but the final text, analysis, and coding must reflect the student’s own understanding. Students are responsible for verifying all facts, debugging code, and ensuring originality; failure to comply may result in assignment or course penalties per university policy. Please reach out to me if you have any questions.

Religious Accommodations

Please review by accessing the link in the Classroom Procedures, Policies, and Student Resources module in the Canvas course.

Student Resources

Title IX

Please review by accessing the link in the Classroom Procedures, Policies, and Student Resources module in the Canvas course.

Sexual Assault Resources

Please review by accessing the link in the Classroom Procedures, Policies, and Student Resources module in the Canvas course.

Assignment Directions

Please read the directions carefully on each homework and project assignment webpage:

- Homework 1
- Homework 2
- Homework 3
- Homework 4
- Midterm Project
- Final Project

Course Calendar

The Fall 2026 University of Cincinnati semester runs from August 24 through December 12, 2026. Full-session classes run from August 24 through December 5, 2026, and exam week runs from December 6 through December 12, 2026.

University Calendar Dates

Date	University Calendar Note	Course Impact
Monday, September 7	Labor Day	UC is closed; no Tuesday class meeting is affected.
Monday, October 12	Reading Day	Regular classes are suspended; no Tuesday class meeting is affected.
Tuesday, November 3	Reading Day	No class meeting.
Wednesday, November 11	Veteran’s Day	UC is closed; no Tuesday class meeting is affected.
Thursday, November 26-Sunday, November 29	Thanksgiving Weekend	UC is closed; no Tuesday class meeting is affected.
Saturday, December 5	Fall classes end	Final instructional week ends.

Date	University Calendar Note	Course Impact
Sunday, December 6-Saturday, December 12	Exam Week	No regular class meeting.

Class	Date	Assignment/Due Date	Topic	Reading	Note
1	Tuesday, August 25	HW 1 assigned	Intro to Multivariate	Link	Asynchronous/9:30-10:20: Watch video lecture; In person/10:30-12:20: Discussion/lab
2	Tuesday, September 1	Midterm Project assigned	PCA/EFA	Link	Asynchronous/9:30-10:20: Watch video lecture; In person/10:30-12:20: Discussion/lab
3	Tuesday, September 8		EFA		Asynchronous/9:30-10:20: Watch video lecture; Teams/10:30-12:20: Discussion/lab
4	Tuesday, September 15	HW 1 due at midnight	EFA/Regression	Link	Asynchronous/9:30-10:20: Watch video lecture; Teams/10:30-12:20: Discussion/lab
5	Tuesday, September 22	Midterm Project proposal; HW 2 assigned	Adv Regression	Link	Asynchronous/9:30-10:20: Watch video lecture; In person/10:30-12:20: Discussion/lab
6	Tuesday, September 29		Work Day		Asynchronous/9:30-10:20: Watch video lecture; Teams/10:30-12:20: Discussion/lab
7	Tuesday, October 6	Midterm Project presentation	Presentation Day		Asynchronous/9:30-10:20: Watch video lecture; In person/10:30-12:20: Discussion/lab

Class	Date	Assignment/Due Date	Topic	Reading	Note
8	Tuesday, October 13	Midterm Project paper; HW 3 assigned	Intro to SEM	Link	Asynchronous/9:30-10:20: Watch video lecture; Teams/10:30-12:20: Discussion/lab
9	Tuesday, October 20	Final Project assigned; HW 2 due at midnight	SEM		Asynchronous/9:30-10:20: Watch video lecture; In person/10:30-12:20: Discussion/lab
10	Tuesday, October 27		SEM, CFA, and PA	Link	Asynchronous/9:30-10:20: Watch video lecture; Teams/10:30-12:20: Discussion/lab
11	Tuesday, November 3		Reading Day		No class: UC reading day
12	Tuesday, November 10	HW 3 due at midnight; HW 4 assigned	CART	Link	Asynchronous/9:30-10:20: Watch video lecture; Teams/10:30-12:20: Discussion/lab
13	Tuesday, November 17	Final Project proposal due	ML and LLMs	Link	Asynchronous/9:30-10:20: Watch video lecture; Teams/10:30-12:20: Discussion/lab
14	Tuesday, November 24	HW 4 due at midnight	Work Day		Asynchronous/9:30-10:20: Watch video lecture; Teams/10:30-12:20: Discussion/lab
15	Tuesday, December 1	Final Project presentation	Presentation Day		Asynchronous/9:30-10:20: Watch video lecture; In person/10:30-12:20: Discussion/lab
16	Tuesday, December 8	Final Project paper	Finals Week		No regular class meeting

Changes to Course Syllabus

This syllabus is merely the instructor’s projection of the expected progress through this course. Circumstances may arise that warrant its revision. Should these circumstances arise, this course syllabus may be subject to change. Students will be notified in a timely manner. The instructor also reserves the right to introduce materials not covered in the textbook. For any situation or topic not covered in the syllabus, please refer to the Nursing Student Handbook.